LUMYVUE V6 - the model, explained in full

A model-only document for readers (human or AI) who need to understand exactly what the final V6 model is, how it was built and trained, how it makes a decision, what it was evaluated on and how far the evidence goes. Run: open_run02 (finished 2026-11-02); deployed in the app on branch v6-integration; status 2026-11-06. Every number comes from the run's saved files; nothing is rounded up or tuned for this document.

1. In one paragraph

V6 is a decision system for three facial skin conditions - Acne, Hyperpigmentation and Eczema - applied to one photograph at a time. It is not a single neural network but a small system: (a) a whole-photo MobileNetV2, (b) three YOLOv8 lesion detectors, (c) optionally a region-level MobileNetV2 that reads crops chosen by the detectors, and (d) a small trained fusion network that combines all of that evidence into one probability per condition. The probabilities are calibrated, compared with per-condition thresholds, and checked by trust rules (conflict, out-of-distribution, photo quality). The output is one unified, reviewable result per photograph. It replaced V5, a single 4-class MobileNetV2 classifier whose answer could disagree with the lesion detectors. On the fair comparison set V6 matches V5 accuracy; it does not beat it in a statistically clear way (section 9).

2. The task

- **Multi-label, not multi-class.** One photograph can show several conditions at once. Conditions (fixed order): Acne, Hyperpigmentation, Eczema.
- **Normal is derived, never predicted.** A photograph is 'no findings supported' only when all three conditions are Absent, every detector worked, no flag was raised and the photo quality was assessed. V5, by contrast, had a 4th 'Normal' class.
- **Unknown labels are allowed.** A condition can be unknown for a photograph (not verified). Unknown entries are masked out of the loss and out of the metrics instead of being treated as negative.
- **Intended output:** per condition Present / Not found / Uncertain, a set of review flags, and detector boxes as possible locations. It is a screening aid; every result waits for dermatologist review.

3. Data

3.1 Source images

Source folders by class label (images before quality and duplicate filtering): Acne 3,000, Hyperpigmentation 3,000, Eczema 1,063, Normal 3,000 (10,063 in total). Image types per source class include close-up facial, full face, lesion-region facial and partial face. Photos flagged as rejected / manual-review by the facial QA, exact duplicates beyond the first copy, and non-facial auxiliary images were excluded; 9,789 images remain across the four splits below.

3.2 Splits and leakage control

Splits are made on **leakage groups**, not single images: images linked by identical file hash (274 links), shared filename stem (3,062 links) or near-duplicate perceptual hash (1,635 links, Hamming distance at most 3) always land in the same split. Split fractions 60 / 10 / 15 / 15 with seed 2026. Each split has a purpose:

Split	Images	Fully verified photos	Present (Acne / Hyper. / Eczema)	Used for
train	5,873	889	176 / 261 / 243	Training the Global CNN, ROI CNN (and the earlier detectors' data are separate)
validation	979	591	173 / 233 / 55	Choosing between experiments, early stopping, V5-vs-V6 fair set
fusion	1,469	347	111 / 124 / 33	Training the fusion network by cross-fitting, calibration, thresholds, bands, OOD, development metrics. Separate so that fusion never learns from images the CNNs trained on

Split	Images	Fully verified photos	Present (Acne / Hyper. / Eczema)	Used for
test (locked)	1,468	0	0 / 0 / 0	Never opened. It has no verified labels, so it cannot yet serve as evidence

- **Label quality:** a photograph's folder name is only a weak positive for that condition in training (never a negative for the others, never used for evaluation). Evaluation uses only 'verified' labels. Verified photographs are a small fraction of all images (889 of 5,873 in train).
- **Label noise check:** 239 suspicious photographs were re-reviewed; only 12 changed, so label noise is not believed to be the main limit.
- **V5 exposure:** V5 had trained on many validation photographs. The 'fair set' (132 photographs V5 never saw) removes this advantage: 47 hyperpigmentation-positive, 26 all-negative, 18 eczema-positive.

4. Component 1 - the whole-photo CNN (Global model)

Item	Value
Backbone	MobileNetV2, 224x224x3 input. Selected model 'E_v5_backbone_multilabel3': initialised from V5's backbone weights (the alternative D_multilabel3 started from ImageNet weights).
Head	Global average pooling -> Dense(128, ReLU, L2 1e-4) named 'embedding' -> Dense(3) logits (sigmoid gives one probability per condition). Dropout 0.3.
Loss	Masked binary cross-entropy on logits: unknown labels masked out; per-condition positive weight, clipped to 1-8, to offset class imbalance.
Training	Phase A: backbone frozen, learning rate 1e-3, up to 12 epochs. Phase B: fine-tune from layer 100, learning rate 1e-5, up to 20 epochs. Adam; early stopping patience 6; LR reduced x0.5 after 3 stalled epochs (min 1e-7); batch 8; seeds 42, 7, 123 tried; deterministic.
Augmentation	Horizontal flip, rotation 10 deg, translation 5%, zoom -30%..+10%, brightness and contrast 8%, random scale-crop (p 0.5). Hard-example up-weighting (x3, at most 25%) exists only for the hard-example experiments.
Preprocessing (identical in the app)	BGR to RGB, bilinear stretch to 224x224 (not letterboxed), (pixel / 127.5) - 1.
Outputs used downstream	128-number embedding and 3 logits. The fusion consumes logit(clip(sigmoid(z), 1e-4, 1-1e-4)), i.e. z clipped to +/-9.21.
Selection	Chosen by validation macro average-precision: E 0.863 vs D 0.796 (95% CI of the difference 0.045 to 0.091); E showed no per-condition F1 regression against V5, D regressed on Acne and Eczema. Selected seed 42.

Validation performance at threshold 0.5 (verified photographs, n about 594 per condition), V5 baseline vs the selected Global model:

Condition	V5 F1	V5 AP	Global E F1	Global E AP	Global E recall	Global E precision
Acne	0.860	0.875	0.865	0.925	0.855	0.876
Hyperpigmentation	0.779	0.874	0.793	0.912	0.708	0.902
Eczema	0.689	0.651	0.690	0.753	0.709	0.672

These validation numbers include photographs V5 had trained on, so they flatter V5; the fair set in section 9 corrects for that.

5. Component 2 - the three YOLOv8 lesion detectors

- **One specialist per condition**, each an ensemble of three models trained with different seeds (Acne: YOLOv8s base, seeds 42/7/123, input 800 px; Hyperpigmentation: three seeds, 800 px; Eczema: YOLOv8n base, seeds 42/43/44, 640 px). They were trained before V6 and are shared with V5; V6 does not retrain them.
- **Inference replicates the app exactly:** letterbox to the model size (grey 114 padding, bilinear), RGB, divide by 255, NCHW; the output [1, 4+classes, N] holds box centre/size and class scores with no objectness; per-member NMS (IoU 0.6) with every member reporting boxes down to confidence 0.02; boxes from the three members are clustered by confidence-weighted average (IoU at least 0.5); fused confidence = min(1, sum of member confidences / 3), so a patch only one member sees gets a low score.
- **Deployed thresholds** (shown boxes): Acne 0.275, Hyperpigmentation 0.225, Eczema 0.25. V6 additionally keeps a **candidate pool** down to confidence 0.05 (up to 60 boxes per condition) and applies **no face filter**, because the training evidence was built without one.
- A detector that is missing or fails is 'unavailable', never 'no detection'. Absence can only be claimed with a working detector.
- Detector accuracy figures are deliberately not quoted here: the hyperpigmentation detector's saved metrics file does not match the hash of the installed ONNX, so its stored mAP/F1 cannot be attributed to the installed model. Re-measuring it on hand-boxed photographs is an open task.

5.1 Detector evidence vector (21 numbers)

For each condition (Acne, Hyperpigmentation, Eczema), 7 numbers: available (0/1), detected (at least one box above the deployed threshold), confidence of the best deployed box, highest confidence anywhere in the pool, square root of

the best box's relative area, and the box centre x and y as fractions of the frame. A missing detector yields seven zeros including available = 0. Box counts are never used. Values are rounded as in training (confidence 5 decimals, area 6, centre 4).

6. Component 3 - the region branch (ROI MobileNetV2; full variant only)

This is where detector information flows into a second CNN. For each photograph the detectors' boxes select up to 12 regions:

- **Tier 1:** up to 6 deployed boxes, taken round-robin across the three detectors so one chatty detector cannot fill every slot; a box overlapping an already chosen one (IoU above 0.70) is dropped.
- **Tier 2:** lower-confidence pool boxes that are not duplicates, filling the remaining slots.
- **Tier 3:** wider-context copies (60% margin instead of 25%) of the first tier-1 boxes, at most 3.
- **Crop:** expand the box by the margin, make the window square by extending the shorter side, shift it inside the image (never zero-pad), resize to 224x224 (bilinear), scale to [-1, 1]. Regions smaller than 2% of the short side, outside the frame or degenerate are skipped.
- **ROI network:** MobileNetV2 (ImageNet init), 4 classes: Acne, Hyperpigmentation, Eczema, Background; 64-number embedding. Trained on the detectors' actual proposals, labelled from verified region annotations (IoU at least 0.50 to a verified region = that class; a proposal touching no verified region (IoU below 0.10) on a verified-normal photo = Background). Phase A frozen backbone 8 epochs, phase B partial fine-tune 12 epochs; seed 42; frozen-backbone macro-F1 0.727, fine-tuned 0.777 (chosen).
- **Per-region feature (83 numbers):** 64 embedding + 4 class scores + 15 geometry/evidence values (condition one-hot, confidence, relative area, centre, width, height, log aspect ratio, deployed flag, ensemble support fraction, tier flags). A mask marks real regions; a photograph with no regions gives an all-false mask, never a blank crop.

ROI network validation (1,378 region crops): accuracy 0.840, macro-F1 0.777.

Region class	Support	Precision	Recall	F1	PR-AUC
Acne	275	0.752	0.927	0.831	0.919
Hyperpigmentation	132	0.620	0.864	0.722	0.827
Eczema	99	0.579	0.778	0.664	0.661
Background	872	0.986	0.817	0.893	0.982

7. Component 4 - the fusion network

A small trained network that takes everything and outputs 3 logits (one per condition). Inputs: Global embedding (128), Global logits (3), an extra 2-number slot (zeros for the multi-label Global model), the 21 detector numbers, 12 regions x 83 features and a 12-number region mask.

Layer	What it does
Global projection	Dropout 0.3, then Dense(64, ReLU) on the 128-number embedding
Detector branch	21 detector numbers through a modality-dropout layer (rate 0.25 during training: the whole group is zeroed for a random fraction of samples so the network never relies on it)
Region branch	Dense(64, ReLU) per region, then masked attention pooling with 3 heads (one per condition); an all-masked set gives zeros; fraction of occupied region slots is also passed on
Fusion	Concatenate [projected embedding, Global logits, extra slot, detector numbers, pooled regions, region fraction] -> Dense(64, ReLU, L2 1e-4) -> Dropout 0.3 -> Dense(3) logits

- **One architecture, several variants** by hiding input groups (zeros plus mask): **full** (global + detector evidence + regions), **global_yolo** (no regions - the default deployed variant), **global_only** (detector evidence hidden - used only for the conflict flag), **roi_only**, and three 'no X evidence' ablations.
- **Training:** masked BCE (per-condition positive weights), Adam 1e-3, up to 150 epochs, early stopping patience 20, batch 2. Trained on the fusion split only, with **grouped 5-fold cross-fitting**: every fusion-split photograph gets an out-of-fold prediction from a model that never saw it (or any photograph of its leakage group). Final model: seed 42 (a mean of per-seed logits is supported; one seed was run).
- **Why a separate fusion split:** the CNNs and detectors must not have trained on the images the fusion learns from, otherwise their outputs on those images are over-confident and the fusion learns to trust them too much.

8. From scores to a decision

8.1 Calibration and thresholds

Platt scaling per condition on the out-of-fold fusion logits: p = sigmoid(a * z + b), fitted by regularised log-loss (L2 1e-3 pulling a toward 1). Thresholds are chosen on the same out-of-fold predictions by scanning 0.05..0.95 in steps of 0.01 and taking the F1-maximising value. Verdict per condition: Present if p >= threshold + band; Not found if p < threshold - band; otherwise Uncertain. The band is 0.0 (Stage 1 band selected on out-of-fold data to reach at most 10% error on accepted cases with at least 30% coverage: result band 0.0, coverage 0.41, error 0.054, 147 accepted known cases) - marked PROVISIONAL.

Variant	Platt a, b - Acne	Hyperpigmentation	Eczema	Thresholds (A / H / E)
global_yolo (default)	0.730, -0.681	0.762, -0.547	0.536, -1.733	0.42 / 0.37 / 0.27
full (optional)	0.583, -0.807	0.628, -0.298	0.501, -1.662	0.43 / 0.34 / 0.33
global_only (flag only)	0.794, -0.488	0.880, -0.456	0.594, -1.905	0.51 / 0.39 / 0.25

8.2 Review flags

- **cnn_yolo_conflict:** the calibrated global_only probability says present/absent differently from whether the detector marked an area, for a condition whose detector is available. The global_only network never changes a score.
- **near_threshold:** the calibrated probability is within the band of the threshold (inactive while the band is 0.0).
- **ood (out of distribution):** Mahalanobis distance of the 128-number Global embedding to the training distribution (Ledoit-Wolf covariance); flagged above 14.74, the 99th percentile of the training distances. Means 'unlike the training photographs', not 'an unknown condition'.
- **quality:** blocking problems (e.g. no face in live capture, too dark/bright/blurry) ask for a retake; warnings are noted. The quality thresholds are provisional and not validated.

8.3 Overall state

FINDINGS_SUPPORTED / MULTIPLE_FINDINGS_SUPPORTED (a condition Present, all others resolved, no flags); NO_FINDINGS_SUPPORTED (all Absent with working detectors, no flags, quality assessed); UNCERTAIN_REVIEW_REQUIRED (any Uncertain condition, unsupported absence, or flag); RECAPTURE_REQUIRED; OUT_OF_SCOPE_REVIEW_REQUIRED; INFERENCE_ERROR (never a normal-looking result). **Detector-mark rule added at deployment:** if the fusion says Not found but that condition's detector marked an area, the verdict is shown as 'Possible' and 'no findings' is impossible. The detector never overrides a Present.

8.4 What was not deployed

The training notebook also defines a Stage 2 'AI verification' loop that escalates doubtful cases to the region branch (band 0.14, coverage 0.84, error 0.099 on accepted cases). The application does not run Stage 2; a case that would escalate is reported as UNCERTAIN_REVIEW_REQUIRED. Only the full variant uses regions at all, and by default the app runs global_yolo.

9. Results

9.1 Development evaluation (fusion split, out-of-fold)

347 fully verified photographs of the fusion split, predictions out-of-fold. This is development data: calibration, thresholds and band were chosen on it, so it is optimistic and is not independent evidence.

Variant	Condition	Precision	Recall	F1	AP	ROC-AUC
global_yolo	Acne	0.876	0.892	0.884	0.950	0.971
global_yolo	Hyperpigmentation	0.918	0.903	0.911	0.952	0.958
global_yolo	Eczema	0.735	0.758	0.746	0.711	0.915
full	Acne	0.877	0.901	0.889	0.926	0.965
full	Hyperpigmentation	0.910	0.895	0.902	0.952	0.964

Variant	Condition	Precision	Recall	F1	AP	ROC-AUC
full	Eczema	0.778	0.636	0.700	0.720	0.890

Multilabel summary: global_yolo macro-F1 0.846, micro-F1 0.880, exact match 0.862; full macro-F1 0.827, micro-F1 0.875, exact match 0.865. On development data the region branch does not help (Eczema recall drops).

9.2 The fair comparison against V5 (validation photographs V5 never trained on, n = 132)

System	Hyper. missed (of 47)	Clean flagged hyper. (of 26)	F1 acne	F1 hyper.	F1 eczema	Macro-F1
V5 argmax	9	1	0.835	0.658	0.718	0.737
V5, thresholds tuned	7	2	0.836	0.725	0.667	0.743
Global only (re-calibrated)	11	2	0.874	0.711	0.595	0.726
V6 global_yolo (default)	10	2	0.870	0.711	0.632	0.737
V6 full (optional)	8	2	0.862	0.759	0.645	0.755
ROI only	9	1	0.830	0.734	0.629	0.731

'Hyper. missed' = a hyperpigmentation photograph for which the system names no condition at all. Paired cluster-bootstrap 95% intervals against V5: global_yolo macro-F1 +0.001 (-0.084 to +0.074), missed-rate +0.021 (0.000 to +0.076), false-alarm rate +0.038 (-0.081 to +0.185); full macro-F1 +0.019 (-0.086 to +0.112), missed-rate -0.021 (-0.093 to +0.053), false-alarm rate +0.038 (0.000 to +0.136). Every interval includes zero (some touch it at an edge).

9.3 What the evidence supports

- **Pre-registered success criterion not met:** (a) false alarms not better than V5, (b) Eczema F1 more than 0.05 below V5. V6 is not shown to be more accurate than V5.
- V6 ranks eczema better than V5 (AP 0.75 vs 0.66, but on only 18 positives) and loses at the calibrated cutoff. V5 with thresholds tuned on the fusion split already captures most of the gain (macro-F1 0.743).
- The region branch adds nothing reliable over global_yolo; this is why global_yolo is the default.
- Detectors alone catch about 91% of hyperpigmentation photographs on the fair set but flag about 27% of clean photographs - the reason a detector mark that the fusion did not confirm is shown as 'Possible', not as a finding.
- Threshold sweeps were flat: the remaining errors are not a threshold problem.
- What V6 adds is not accuracy but structure: one result instead of two contradicting answers, calibrated probabilities, explicit flags and abstention, auditable sealed records, and detector-evidence maps.

10. Deployment in the application and verification

- Exported to ONNX (global CNN, ROI CNN, three fusion variants) and run in the C# WPF app with Microsoft.ML.OnnxRuntime on CPU. Each model file's SHA-256 is checked against the deployment config at load; a mismatch disables V6.
- C# reimplements the detector post-processing, the 21-number evidence, the region selection and cropping, calibration, thresholds, flags and states. A Python reference that reproduces the training-time detection cache exactly (13 of 13 photographs) generated golden values: C# vs Python detector boxes 270 boxes within 0.0001 px; fusion CPU parity max difference 1.9e-6 with 0 of 1200 decisions flipped; full-variant calibrated probabilities within 1e-6 with 0 of 39 decisions flipped.
- The 'outside tolerance' ONNX-vs-Keras difference seen in the notebook was GPU-versus-CPU noise: CPU ONNX matches CPU Keras to about 5e-5.
- Run time: about 2 s per photograph for the full variant on CPU in tests (global_yolo is faster). Live camera reading runs in a background loop (newest frame wins).
- V5 stays inside the app (feature flag V6Unified; variant switch V6Variant). The V6 decision never averages or votes with V5.
- Each scan stores a sealed JSON (SHA-256 seal) with the full assessment, flags, variant and model hashes, plus the evidence-map images.

11. Heat maps (evidence maps)

V6 maps show detector evidence: a Gaussian bump per detector box (the maximum, not the sum, where bumps overlap), strength = box confidence / 0.70 capped at 1, drawn only inside the face region and coloured per condition. They show where the detectors found evidence, not why the classifier decided. A Grad-CAM through the fusion network was prototyped and rejected because the maps for different conditions were 95-98% identical.

12. Limitations and threats to validity

- **Small verified sets.** 889 / 591 / 347 fully verified photographs (train / validation / fusion); the fair set has 132 photographs (18 with eczema). Confidence intervals are wide.
- **Development, not independent.** Validation was used to choose experiments and stop training; the fusion split to calibrate, set thresholds and band. The locked test split has no labels, so there is no independent test result. The run's own metadata labels it an experimental development run.
- **Weak labels.** Folder names act as weak positives in training; verified labels are limited in number.
- **Single seed** for the ROI network and fusion network (the Global CNN tried three seeds).
- **Hyperpigmentation generalisation** is the weak spot: it works on hand-tested photographs but has no larger measured result on device photographs; the detector accuracy needs re-measuring on hand-boxed photographs.
- **Eczema** has the fewest examples (55 validation, 33 fusion positives) and is not improved by V6.
- **Quality and OOD thresholds are provisional.** There is no clinical validation, no dermatologist review yet and no real-device test set; the system is a research prototype and must not be used for diagnosis.
- **Skin-tone and demographic fairness** were not evaluated.

13. Reproducibility

Item	Value
Run	open_run02, created 2026-11-02; feature schema astra-fusion/1; assessment schema lumyvue-final-assessment/1.1; app archive schema lumyvue-v6-assessment/1.0
Software	Python 3.13, TensorFlow 2.21.0, onnxruntime 1.30.0; deterministic mode; seeds 42 / 7 / 123 (Global), 42 (ROI, fusion)
Model hashes (SHA-256, first 16)	global features 82662184cc526f26; fusion global_yolo f25a741cae5d87b5; fusion global_only fd9ef9a6803f30a9; fusion full 1dbea01249c7...; ROI features 45a65103d89c4c59

Item	Value
Where things are	Training notebooks and sources in v6/ (astra_training_src, run outputs in v6/astra_runs/open_run02); deployment package in v6/deployment; models in SkinDevApp/Model/V6; parity references v6_real_photo_golden.json and v6_full_variant_golden.json

Glossary: AP = average precision (area under the precision-recall curve); F1 = harmonic mean of precision and recall; Platt scaling = fitting a sigmoid to turn raw scores into calibrated probabilities; OOD = out of distribution; ROI = region of interest; NMS = non-maximum suppression; IoU = intersection over union; OOF = out-of-fold; macro-F1 = average of the three per-condition F1 scores.